## PEER REVIEW HISTORY

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## ARTICLE DETAILS

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| **TITLE (PROVISIONAL)** | The Relationship between Hormonal Contraceptives and Sleep among Women of Reproductive Age: A Systematic Review Protocol |
| **AUTHORS** | Ma, Jinhui; Cheng, Megan; Thabane, Lehana; Ma, Caihong; Zhang, Ning; Wang, Qi; Kim, Hyunwoo; Reza, Hameed; Wang, Chunxue; Yao, Xiaomei |

## VERSION 1 – REVIEW

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| **REVIEWER** | de Wit, Anouk<br>University Medical Centre Groningen |
| **REVIEW RETURNED** | 21-Nov-2020 |

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| **GENERAL COMMENTS** | This paper presents the protocol for a systematic review on the effects of oral contraceptives on sleep. Oral contraceptive-associated sleep alterations are complex and involves a broad spectrum of factors. The authors succeeded in planning a comprehensive review of the current literature in order to inform the research field and women about the current state of knowledge on this topic. The manuscript in its current form introduces several interesting planned analyses, however the paper would benefit from revisions addressing the following concerns:<br><br>1. The authors plan to conduct a comprehensive review of the current literature on the effects of oral contraceptives on sleep. However, why do the authors limit their search to oral contraceptives? I think only including oral contraceptives is a pity as a large group of hormonal contraceptives such as implants, rings and hormonal IUDs, that also have systemic effects, are missed.<br>2. The authors state in the introduction: “Findings from this review will provide insights to women choosing between different methods and formulations…” However, I think that if the authors aim for this goal, different statistics are needed as the pairwise meta-analysis planned, will not allow the researchers to compare the effects of different oral contraceptives. I would therefore suggest to authors to conduct a network-meta-analysis. A network meta-analysis of randomized controlled trials makes it possible to estimate the comparative harm of different oral contraceptives using both direct and indirect evidence, and hence to summarize and interpret the wider picture of the evidence for their putative effect on sleep.<br>3. Some information is missing or unclear to me:<br>a. Are studies that included women who were not off oral contraceptive use prior to inclusion allowed in this review?<br>b. The authors plan to include women between 15 and 49 years old. I would propose to be more specific. Are women with |

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| | comorbidities such as PCOS be allowed? Do women need to have normal menstrual cycles? Might they be postmenopausal?<br>c. Also, women included need to use the oral contraceptives to avoid pregnancy (page 10, row 6 to 10). However, why do the authors exclude women who take the oral contraceptives for other reasons like acne or dysmenorrhea?<br>d. What will authors do when articles use different sleep measurements to assess the same outcome? Will priority given to certain methods?<br>e. The authors don't describe at which point the outcome on sleep will preferably assessed. After days? months? years?<br>f. Authors state that non-randomized comparative studies need to include potential confounders. Is there a minimum number of confounders or specific confounders that need to be included? (page 8 row 38 to 41)<br>g. The authors calculate total sleep time as the total amount in bed minus the time to takes to fall asleep plus the time spent awake. Why do the authors add the time spent awake? (page 11 row 15 to 17)<br>h. What do the authors mean with oral contraceptives? Do they include progesterone-only-pills e.g. to be included?<br>4. The authors wrote in the introduction: Women are more likely than men to have sleep problems including …. since changing hormones during the menstrual cycle, pregnancy, and menopause can affect how well a woman sleeps. This implies that hormones solely can explain the gender gap in sleep disorders. This is however a too simplistic statement. I would suggest to tone down this statement and mention that hormones potentially partially can explain the gender gap.<br>5. The authors wrote in the introduction: Start from the mid-lutual phase, steroidal hormone levels decrease... "Steroidal hormones" is too vague. What do the authors mean by "steroidal hormones"? All hormones derived from cholesterol?<br>6. I miss some hypotheses about what sleep outcomes are presumed to be affected. Do the authors have thoughts about what sleep outcomes would be affected? Or all sleep outcomes? |

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| **REVIEWER** | Sahni, Ashima<br>University of Illinois System |
| **REVIEW RETURNED** | 23-Dec-2020 |

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| **GENERAL COMMENTS** | As the protocol is looking at insomnia related outcomes and not OSA or RLS related outcomes of sleep disruption, the title should be reflective of the same. Therefore, I suggest that the title should be changed to " Do Oral contraceptives affect Insomnia among women of reproductive age?" or other causes of sleep disruption should be added to the outcome section of the protocol. |

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| **REVIEWER** | Driver, Helen<br>Queen's University |
| **REVIEW RETURNED** | 29-Dec-2020 |

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| **GENERAL COMMENTS** | The investigators propose that this systematic review will provide insights to help women choose between different methods and formulations of OC, while also informing the development of new progestins for future generations of OC.<br><br>Page 5: The following statement is misleading and not entirely accurate. "Women are more likely than men to have sleep |

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| | problems including insomnia, restless leg syndrome, and sleep apnea since changing hormones during the menstrual cycle, pregnancy, and menopause can affect how well a woman sleeps.13-15" The incidence of sleep apnea is higher in men than in women. While the changing hormonal milieu may influence how "well a woman sleeps", this influence on sleep is not the reason they experience the sleep disorders listed above.<br>Page 5: There is evidence, largely based on subjective studies, that from the mid-luteal phase (about 6 days pre-menstruation), women report more sleep disturbances. More appropriate references are available in this regard than cited. Furthermore, editing of the following statement is advised "Start from the mid-luteal phase, steroidal hormone levels decrease and women are awakened during sleep more often compared with the follicular phase.17 (17. Rivera R, Yacobson I, Grimes D. The mechanism of action of hormonal contraceptives and intrauterine contraceptive devices. Am J Obstet Gynecol [Internet]. 1999 Nov [cited 2020 Jul 20] ;181(5 Pt 1):1263–9. Available from: https://pubmed.ncbi.nlm.nih.gov/10561657/)<br><br>Page 6: There is indication that the route of OC administration may offer potential advantages for depot over oral in the pilot study by Guida et al 2020.<br><br>Search Terms: As outlined on Page 7-8, their search terms covered all words related to birth control, ingredients of oral contraceptives, and sleep. Additional search terms suggested would include terms "menstrual cycle" in particular given that some studies may have compared sleep to normal cycling women, "hormonal contraceptives", "sleep spindles", "PSQI" (Table 1). Limiting the search to studies or systematic reviews published from 1970 to present is warranted to help to ensure that only second and newer generation OCs are included. Searching "sleep" terms in the full text besides the title and abstract is also warranted to capture those studies where sleep was a secondary outcome.<br><br>Limited number of studies: Given the challenges and small number of original studies on human female participants that have investigated this topic, what would the minimum number of studies need to be in order to adequately inform the proposed systematic review? Indeed although recommendations on OC formulation may be limited, through a thorough appraisal of published studies, this systematic review could help to highlight the need for research in this field as well as provide suggestions in study design and implementation. |

**VERSION 1 – AUTHOR RESPONSE**

**Reviewer: 1** (Dr. Anouk de Wit, University Medical Centre Groningen)
This paper presents the protocol for a systematic review on the effects of oral contraceptives on sleep. Oral contraceptive-associated sleep alterations are complex and involves a broad spectrum of factors. The authors succeeded in planning a comprehensive review of the current literature in order to inform the research field and women about the current state of knowledge on this topic. The manuscript in its current form introduces several interesting planned analyses, however the paper would benefit from revisions addressing the following concerns:

1. *The authors plan to conduct a comprehensive review of the current literature on the effects of oral contraceptives on sleep. However, why do the authors limit their search to oral contraceptives? I think only including oral contraceptives is a pity as a large group of hormonal contraceptives such as implants, rings and hormonal IUDs, that also have systemic effects, are missed.*

Response

We greatly thank Dr. de Wit for her insightful comments and suggestions. We agree that including not only oral contraceptives but also other hormonal contraceptives such as implants, rings and hormonal IUDs will make this systematic review more comprehensive. We have added a paragraph regarding the other hormonal contraceptives and revised the objective of the systematic review as follows:

> Other hormonal contraceptives have been widely adopted as well and used by over 97 million women in the world. These hormonal contraceptives have different dosage, formulation, mechanism of action, and applicable groups compared to OC. Unlike oral contraceptives which are taken consistently and provide a stable dosage of the estrogen and progestin components, other hormonal contraceptives have the potential to provide lower dose of progestin and estrogen. It has been shown that hormonal IUDs release the levonorgestrel directly into the uterus, only a small amount is absorbed into the rest of the body, i.e. its effects are mostly paracrine rather then systemic. An investigation of Nexplanon, an implant contraceptive lasting 3 years, found progestin absorption at 60 mg/day after 3 months and 30mg/day after 6 months. (see page 5 in the manuscript)
>
> Hence, the objective of this systematic review is to compile and elucidate the association between hormonal contraceptives, especially OC, use and sleep (including the quality of sleep, sleep disorders, and duration of sleep). (see page 7 in the manuscript)

2. *The authors state in the introduction: "Findings from this review will provide insights to women choosing between different methods and formulations…" However, I think that if the authors aim for this goal, different statistics are needed as the pairwise meta-analysis planned, will not allow the researchers to compare the effects of different oral contraceptives. I would therefore suggest to authors to conduct a network-meta-analysis. A network meta-analysis of randomized controlled trials makes it possible to estimate the comparative harm of different oral contraceptives using both direct and indirect evidence, and hence to summarize and interpret the wider picture of the evidence for their putative effect on sleep.*

Response

We agree with Dr. de Wit's suggestion about conducting a network meta-analysis and have added the following under the ***Data synthesis*** subsection in the METHODS section:

> As long as studies included in the systematic review form a connected network, a network meta-analysis using both direct and indirect evidence will be conducted to estimate the comparative harm of different contraceptives. Results from the network meta-analysis will allow us to summarize and interpret a wider picture of the evidence for the putative effect of hormonal contraceptives on sleep. (see page 14 in the manuscript)

3. *Are studies that included women who were not off oral contraceptive use prior to inclusion allowed in this review?*

Response

Many thanks for raising this good point. For studies with women in the control group using non-hormonal contraceptives or naturally cycling, they will be included in the systematic review only if participants in the control group have maintained not using any hormonal contraceptives for at least three months. To make this clearer, we have revised the ***Study Selection and Screening*** section as follows:

> Participants or subgroups of participants take second generation or more recent OC or other hormonal contraceptives, regardless of its brand, dose, and frequency, to avoid pregnancy, as long as their hormonal contraceptive use pattern is sustained for at least three months. (see page 9 in the manuscript)

> Included studies must have comparison groups of women who are using (i) non-hormonal contraceptives; (ii) naturally cycling (i.e., not use any contraceptive methods); (iii) same agent as the exposure but in different dose, frequency, and duration; and (iv) second generation or more recent OC or other contraceptives but different from the exposure. Since recent usage or cessation of hormonal contraceptive use may cause hormonal oscillations, we will include only studies with participants keeping their contraceptive use pattern (either using or not using) for at least three months to avoid potentially based results. (see page 10 in the manuscript)

4. *The authors plan to include women between 15 and 49 years old. I would propose to be more specific. Are women with comorbidities such as PCOS be allowed? Do women need to have normal menstrual cycles? Might they be postmenopausal?*

Response

Studies focusing on women with any comorbidities or conditions, such as PCOS, hormone disorder, acne, perimenopause, postmenopause, depression etc., will not be included in this review since these comorbidities may be associated with sleep problems as well, which will not allow us to elucidate the association between hormonal contraceptives and sleep.

5. *Also, women included need to use the oral contraceptives to avoid pregnancy (page 10, row 6 to 10). However, why do the authors exclude women who take the oral contraceptives for other reasons like acne or dysmenorrhea?*

Response

We chose to keep strict eligibility criteria to ensure that participants in this review would be as representative of the general population as possible. We exclude studies that focus solely on women with certain conditions such as acne or dysmenorrhea since they are not necessarily representative of all healthy women and these underlying conditions are associated with sleep problems. For instance, studies of individuals with acne have shown they had a reduction in subjective sleep and an increase in acne severity was associated with an increased fatigue upon waking (1). Additionally, the hormonal profile of women with acne is often different from the general population. In a study of 120 women with adult acne, 18.33% had hyperandrogenemia, 13.46% had increased testosterone, 76.92% had increased 17-hydroxyprogesterone and 40.38% experienced irregular menstrual cycles. (2)

(1) Schrom KP, Ahsanuddin S, Baechtold M, Tripathi R, Ramser A, Baron E. Acne Severity and Sleep Quality in Adults. Clocks Sleep. 2019 Dec 6;1(4):510–6.

(2) Sardana K, Bansal P, Sharma LK, Garga UC, Vats G. A study comparing the clinical and hormonal profile of late onset and persistent acne in adult females. Int J Dermatol. 2020 Apr;59(4):428–33.

We presented the above rationales in exclusion criteria section as follows:

> The following studies will be excluded from this systematic review: (1) studies only recruited women with specific comorbidities or conditions, such as hormonal disorder, acne, perimenopause, postmenopause, dysmenorrhea, depression etc., since these conditions may be associated with sleep problems and will not allow us to elucidate the association between hormonal contraceptives and sleep. (see page 12 in the manuscript)

6. *What will authors do when articles use different sleep measurements to assess the same outcome? Will priority given to certain methods?*

Response

Given there are many measurements to assess sleep outcomes and we will include any eligible studies regardless of what measurements are used to assess sleep quality, sleep duration, and sleep disorder outcomes, it is very likely we will have eligible articles using different sleep measures to assess the same outcome. In this case, we will qualitatively summarize the data from these studies for this outcome rather than performing a mate-analysis. We have emphasized this in the ***Data synthesis*** section as follows:

> For sleep measures investigated by two or more studies, if there are no clinical heterogeneity (e.g., different age or intervention) and methodological heterogeneity (e.g., different measurement tools), statistical heterogeneity will be assessed using Q statistic with a p value < 0.1 indicating for significant heterogeneity…. If heterogeneity level is low or medium, we will conduct random-effects meta-analyses to estimate the magnitude of the overall effect of OC, with 95% confidence interval depicting the uncertainty around the estimate. (see page 14 in the manuscript)

7. *The authors don't describe at which point the outcome on sleep will preferably assessed. After days? months? years?*

Response

Many thanks for raising this good point. We have added the following sentence in the outcome session:

> Given we will only include studies with participants keeping their contraceptive use pattern for at least three months, the sleep outcomes from the eligible studies will be summarized after participants keeping their contraceptive use pattern for at least three months as well. (see page 12 in the manuscript)

8. *Authors state that non-randomized comparative studies need to include potential confounders. Is there a minimum number of confounders or specific confounders that need to be included? (page 8 row 38 to 41)*

Response

According to our literature search, age, race, and socioeconomic status are potential confounders that may cause potential bias when estimating the association between hormonal contraceptives and sleep in non-randomized comparative studies. Therefore, we require the included non-randomized comparative studies take into account the effect of potential confounders using methods such as multivariable analysis, propensity-score matching, or showing no statistically significant differences in baseline characteristics between participants in comparison groups. However, we do not want to specify a minimum number of confounders or specific confounders in the protocol since different studies may need to account for different confounders depending on their design and study sample.

9. *The authors calculate total sleep time as the total amount in bed minus the time to takes to fall asleep plus the time spent awake. Why do the authors add the time spent awake? (page 11 row 15 to 17)*

Response

Many thanks for this comment and we have rephrased the sentence as follows:

> The total sleep time is equal to the total amount of time spent in bed (in minutes) minus the sum of time it takes to fall asleep plus and the time spent awake throughout the night. (see page 12 in the manuscript)

10. *What do the authors mean with oral contraceptives? Do they include progesterone-only-pills e.g. to be included?*

Response

In this systematic review, we intend to include any oral contraceptives: combined OC containing both progesterone and estrogen and progesterone-only pills. But we will exclude studies investigating emergency contraception as we described in the Exclusion criteria section in page 12. In the revised manuscript, we included other hormonal contraceptives as suggested and updated the manuscript as follows:

> Participants or subgroups of participants take second generation or more recent OC or other hormonal contraceptives, regardless of its brand, dose, and frequency, and duration, containing both containing both progesterone and estrogen or progesterone-only, to avoid pregnancy, as long as their hormonal contraceptive use pattern is sustained for at least three months. (see page 9 in the manuscript)

11. *The authors wrote in the introduction: Women are more likely than men to have sleep problems including …. since changing hormones during the menstrual cycle, pregnancy, and menopause can affect how well a woman sleeps. This implies that hormones solely can explain the gender gap in sleep disorders. This is however a too simplistic statement. I would suggest to tone down this statement and mention that hormones potentially partially can explain the gender gap.*

Response
We revised this sentence as follows:

> Women are more likely than men to have sleep problems including insomnia and, restless leg syndrome. The exact etiology of these sleep disruptions is unknown but hormonal fluctuations since changing hormones during the menstrual cycle, pregnancy, and menopause have been shown to potentially affect how well a woman sleeps. (see page 6 in the manuscript)

12. *The authors wrote in the introduction: Start from the mid-lutual phase, steroidal hormone levels decrease... "Steroidal hormones" is too vague. What do the authors mean by "steroidal hormones"? All hormones derived from cholesterol?*

Response

We agree it is too vague with using "steroidal hormones" here. We made the following revision in the manuscript:

During the luteal phase, progesterone concentrations increase and estrogen levels are high up to the mid-phase. Starting from the mid-luteal phase, both estrogen and progesterone levels decrease. Sleep disturbances are most commonly reported at the end of the luteal phase and early follicular phase. (see page 6 in the manuscript)

13. *I miss some hypotheses about what sleep outcomes are presumed to be affected. Do the authors have thoughts about what sleep outcomes would be affected? Or all sleep outcomes?*

Response

we have added the hypothesis in the introduction section as follows:

Our hypothesis is that OC and other hormonal contraceptives are associated with a decreased quality of sleep and increased sleep duration in women of reproductive age. (see page 7)

**Reviewer: 2** (Dr. Ashima Sahni, University of Illinois System)

1. *As the protocol is looking at insomnia related outcomes and not OSA or RLS related outcomes of sleep disruption, the title should be reflective of the same. Therefore, I suggest that the title should be changed to " Do Oral contraceptives affect Insomnia among women of reproductive age?" or other causes of sleep disruption should be added to the outcome section of the protocol.*

Response

We thank Dr. Sahni for her valuable comment, and the title has been revised accordingly as "The Relationship between Hormonal Contraceptives and Insomnia among Women of Reproductive Age: A Systematic Review Protocol" (see page 1 in the manuscript).

**Reviewer: 3** (Dr. Helen Driver, Queen's University)

The investigators propose that this systematic review will provide insights to help women choose between different methods and formulations of OC, while also informing the development of new progestins for future generations of OC.

1. *Page 5: The following statement is misleading and not entirely accurate. "Women are more likely than men to have sleep problems including insomnia, restless leg syndrome, and sleep apnea since changing hormones during the menstrual cycle, pregnancy, and menopause can affect how well a woman sleeps.13-15" The incidence of sleep apnea is higher in men than in women. While the changing hormonal milieu may influence how "well a woman sleeps", this influence on sleep is not the reason they experience the sleep disorders listed above.*

   *Page 5: There is evidence, largely based on subjective studies, that from the mid-luteal phase (about 6 days pre-menstruation), women report more sleep disturbances. More appropriate references are available in this regard than cited. Furthermore, editing of the following statement is advised "Start from the mid-luteal phase, steroidal hormone levels decrease and women are awakened during sleep more often compared with the follicular phase.17 (17. Rivera R, Yacobson I, Grimes D. The mechanism of action of hormonal contraceptives and intrauterine contraceptive devices. Am J Obstet Gynecol [Internet]. 1999 Nov [cited 2020 Jul 20] ;181(5 Pt 1):1263–9. Available from: https://pubmed.ncbi.nlm.nih.gov/10561657/)*

Response

We greatly thank Dr. Driver for her insightful comments and suggestions. We have revised that sentence on page 6.

> Women are more likely than men to have sleep problems including insomnia and restless leg syndrome. The exact etiology of these sleep disruptions is unknown but hormonal fluctuations during the menstrual cycle, pregnancy, and menopause have been shown to potentially affect how well a woman sleeps. (see page 6 in the manuscript)

> During the luteal phase, progesterone concentrations increase and estrogen levels are high up to the mid-phase. Starting from the mid-luteal phase, both estrogen and progesterone levels decrease. Sleep disturbances are most commonly reported at the end of the luteal phase and early follicular phase. (see page 6 in the manuscript)

2. *Page 6: There is indication that the route of OC administration may offer potential advantages for depot over oral in the pilot study by Guida et al 2020.*

Response

We greatly thank Dr. Driver for pointing this out. We have revised it as follows:

> For instance, a pilot study published in 2020 by Guida et al found significantly increased sleep duration for users of oral progestins, oral gestodene and 13.5 mg levonorgestrel intrauterine systems, compared with naturally cycling women (women not currently using any hormonal contraceptive), and decreased sleep latency and better sleep quality for users of depot-administered contraceptives. (see page 7 in the manuscript)

3. *Search Terms: As outlined on Page 7-8, their search terms covered all words related to birth control, ingredients of oral contraceptives, and sleep. Additional search terms suggested would include terms "menstrual cycle" in particular given that some studies may have compared sleep to normal cycling women, "hormonal contraceptives", "sleep spindles", "PSQI" (Table 1). Limiting the search to studies or systematic reviews published from 1970 to present is warranted to help to ensure that only second and newer generation OCs are included. Searching "sleep" terms in the full text besides the title and abstract is also warranted to capture those studies where sleep was a secondary outcome.*

Response

We greatly thank Dr. Driver for her suggestion. We have added the suggested search terms in the updated manuscript on pages 8 and in Table 1 on pages 17 to 24.

The term of "sleep*.mp." includes all the terms with sleep and sleep as a prefix in title, abstract, original title, name of substance word, subject heading word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier in Medline; and in title, abstract, subject headings, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword in Embase. In Table 1, we have changed ".mp." to ".af." (.af. = all field) to search the terms with sleep and sleep as a prefix in more fields in Medline and Embase.

4. *Limited number of studies: Given the challenges and small number of original studies on human female participants that have investigated this topic, what would the minimum number of studies*

*need to be in order to adequately inform the proposed systematic review? Indeed although recommendations on OC formulation may be limited, through a thorough appraisal of published studies, this systematic review could help to highlight the need for research in this field as well as provide suggestions in study design and implementation.*

Response

There is no minimum number of eligible studies required for a systematic review. However, to perform a meta-analysis, at least two studies with low heterogeneity are required. In the INTRODUCTION section, we cited four original studies which investigated the association between hormonal contraceptives and sleep. The findings from the four studies are inconsistent, which motived us to conduct a systematic review on this issue. Therefore, we believe at least four studies will be included in the systematic review. Though small number of original studies are available on this topic and the included studies may not support a meta-analysis, we believe through a thorough appraisal of published studies, this systematic review could help to highlight the need for research in this field as well as provide suggestions in study design and implementation.

**VERSION 2 – REVIEW**

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| **REVIEWER** | de Wit, Anouk<br>University Medical Centre Groningen |
| **REVIEW RETURNED** | 26-Mar-2021 |

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| **GENERAL COMMENTS** | I appreciate that the authors made revisions according to my comments. Most of my points were well addressed (4,6,8,9,10,11,12,13).<br><br>However, I still have some questions about the other comments.<br><br>1. Reply to response to comment 1:<br>Please consistently use hormonal contraceptive throughout the manuscript.<br>Just two examples;<br>- Below the header "Strengths and limitations of this study" (page 2), oral contraceptives are mentioned, but not hormonal contraceptives. The main goal was to study hormonal contraceptives if I understood that correctly, so why mention oral contraceptives here?<br>- Below the header "Study selection and screening" (page 9): ".....(iii) studies investigate second generation or more recent OCs; and (iv) studies investigate the association between OC use and sleep-related outcomes, either as the primary or secondary objective."<br>Why use OC here and not hormonal contraceptives?<br>This inconsistent use of hormonal contraceptives versus oral contraceptives was very confusing for me as a reader. I would suggest to use hormonal contraceptives because that matches with the title and the goal of the study.<br><br>2. Reply to response to comment 2:<br>Please be more specific. As this is a method paper, I would expect to read how you will conduct such analyses. For example:<br>- A Bayesian of frequentist approach?<br>- How will you assess convergence?<br>- How will you assess heterogeneity?<br>- etc etc. |

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| | 3. Reply to response to comment 3:<br>Its still not clear to me what the following means: "we will include only studies with participants keeping their contraceptive use pattern (either using or not using) for at least three months to avoid potentially based results"<br>- Do you mean that participants need to be free of hormonal contraceptives at the initiation of the study?<br>- Or are they included if they have at least been on the drug (or not on the drug) for at least three months?<br>- Or do you mean that participants will only be examined if they have used the experimental or control drug for at least 3 three months after the initiation of the study? If so, what will that mean for the analyses? Will you exclude women who dropped out? Wouldn't that bias the results (intention-to-treat versus complete cases analyses)?<br><br>4. Reply to response to comment 5:<br>"We chose to keep strict eligibility criteria to ensure that participants in this review would be as representative of the general population as possible."<br>Please keep in mind that using strict eligibility criteria will reduce the generalizability of the findings. Syndromes like dysmenorrhea and acne are very common and good reasons to initiate OC use. These women therefore are members of the general population using OCs. Excluding these women will reduce the generalizability of the findings. I leave this to the authors, but you may want to consider to include these women and exclude them in a sensitivity analyses to see whether the findings are robust.<br><br>5. Reply to response to comment 7:<br>Please be more specific. What if multiple data points are available? Will the closest datapoint to 3 months be used?<br><br>6. Last comment: the introduction has almost 1300 words. I would consider to shorten this and use the words to more precisely describe the methodology. |

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| **REVIEWER** | Driver, Helen<br>Queen's University |
| **REVIEW RETURNED** | 05-Apr-2021 |

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| **GENERAL COMMENTS** | The authors have addressed my concerns. Thank you. However, one additional point is that they changed the title in response to another reviewers suggestion. In my opinion, the title should not replace "sleep" with "insomnia", which narrows the scope of the review, and would be inaccurate. Insomnia is not the same as decreased sleep quality. The title should remain "Do Oral Contraceptives (OC) Affect Sleep among Women of Reproductive Age? A Systematic Review Protocol" |

**VERSION 2 – AUTHOR RESPONSE**

**Reviewer 1:**

***Comment 1: Please consistently use hormonal contraceptive throughout the manuscript. Just two examples;***

***- Below the header "Strengths and limitations of this study" (page 2), oral contraceptives are mentioned, but not hormonal contraceptives. The main goal was to study hormonal contraceptives if I understood that correctly, so why mention oral contraceptives here?***
***- Below the header "Study selection and screening" (page 9):".....(iii) studies investigate second generation or more recent OCs; and (iv) studies investigate the association between OC use and sleep-related outcomes, either as the primary or secondary objective." Why use OC here and not hormonal contraceptives? This inconsistent use of hormonal contraceptives versus oral contraceptives was very confusing for me as a reader. I would suggest to use hormonal contraceptives because that matches with the title and the goal of the study.***

Response: Thank you for the careful review of our manuscript. We carefully reviewed all mentions of oral contraceptive in the manuscript and replaced with hormonal contraceptives where appropriate.

***Comment 2: Please be more specific. As this is a method paper, I would expect to read how you will conduct such analyses. For example:***

***- A Bayesian of frequentist approach?***

***- How will you assess convergence?***

***- How will you assess heterogeneity?,***

***- etc etc.***

Response: We will use the frequentist approach to obtain the pooled estimate of the effect of hormonal contraceptives on sleep. We revised the Data Synthesis section and provided more details in the manuscript (see page 16).

***Comment 3: Its still not clear to me what the following means: "we will include only studies with participants keeping their contraceptive use pattern (either using or not using) for at least three months to avoid potentially based results"***

***- Do you mean that participants need to be free of hormonal contraceptives at the initiation of the study?***

***- Or are they included if they have at least been on the drug (or not on the drug) for at least three months?***

***- Or do you mean that participants will only be examined if they have used the experimental or control drug for at least 3 three months after the initiation of the study? If so, what will that mean for the analyses? Will you exclude women who dropped out? Wouldn't that bias the results (intention-to-treat versus complete cases analyses)?***

Response: Thanks for pointing this out. We revised the study eligibility criteria for exposure and comparator as follows (please also see page 10 in the revised manuscript):

*Exposure*

*Participants who take second generation or more recent OC or other hormonal contraceptives containing both progesterone and estrogen or progesterone only, regardless of its brand, dose, and frequency, to avoid pregnancy, as long as their hormonal contraceptive use pattern is sustained for at least three months by the time sleep outcomes are assessed. The first three months of hormonal contraception use or discontinuation of hormonal contraception are excluded to account for altered menstruation that occurs following discontinuation of hormonal contraception and significant mood changes seen with initiation of hormonal contraception.*[31-32]

*Comparator*

*Included studies must have comparison groups of women who are using (i) non-hormonal contraceptives; (ii) naturally cycling (i.e., not use any contraceptive methods) or placebo; (iii) same agent as the exposure but in different dose, frequency, and duration; and (iv) second generation or more recent OC or other hormonal contraceptives different from the exposure. Since recent usage or cessation of hormonal contraceptive use may cause hormonal oscillations, we will only include studies with participants keeping their contraceptive use pattern (either using different hormonal contraceptives from the exposure or placebo, or not using any hormonal contraceptive) for at least three months by the time sleep outcomes are assessed.*

We agree that women who drop out the study may cause potential bias if not handled appropriately in the included studies. We will conduct bias assessment for each included randomized controlled trial or non-randomized comparative study. One of the domain to be assessed is bias due to missing data (see page 14 in the revised manuscript). Risk of bias assessments will be incorporated into synthesis by performing a sensitivity analysis (i.e, limiting to studies at lowest risk of bias in a secondary analysis) (see page 15 in the revised manuscript).

***Comment 5: "We chose to keep strict eligibility criteria to ensure that participants in this review would be as representative of the general population as possible." Please keep in mind that using strict eligibility criteria will reduce the generalizability of the findings. Syndromes like dysmenorrhea and acne are very common and good reasons to initiate OC use. These women therefore are members of the general population using OCs. Excluding these women will reduce the generalizability of the findings. I leave this to the authors, but you may want to consider to include these women and exclude them in a sensitivity analyses to see whether the findings are robust.***

Response: We are sorry for the confusion. The eligibility criteria of our systematic review is actually broad in the sense of the multitudes of study designs, contraception formulations and outcomes measured. By “strict”, we mean that we are choosing to exclude papers specific to populations with

syndromes such as dysmenorrhea and acne to keep the results generalizable. For example, the paper “A randomized comparative trial of a combined oral contraceptive and azelaic acid to assess their effect on sleep quality in adult female acne patients” by Albuquerque 2015 (link: https://pubmed.ncbi.nlm.nih.gov/26472098/) focusses exclusively on women with acne would be excluded from our review. We believe sleep changes experienced by women using hormonal contraception are due to hormonal changes in the brain, due to the significant hormonal differences seen in women with dysmenorrhea or acne, papers focussed solely on these women will be excluded.

***Comment 7: Please be more specific. What if multiple data points are available? Will the closest datapoint to 3 months be used?***

Response: We will report any sleep-related outcomes assessed at any time points after keeping their contraceptive use pattern for at least three months. If sleep outcomes are measured at multiple time points in a study, the measure done at the similar time point as the other studies will be used in the primary meta-analysis. The outcome measured at other time points will be considered in the subgroup analysis by duration of hormonal contraceptives. (please see page 16 in the revised manuscript).

***Comment 8: the introduction has almost 1300 words. I would consider to shorten this and use the words to more precisely describe the methodology.***

Response: The introduction has been amended and shortened to 1000 words.

**Reviewer: 3**

***The authors have addressed my concerns. Thank you. However, one additional point is that they changed the title in response to another reviewers suggestion. In my opinion, the title should not replace "sleep" with "insomnia", which narrows the scope of the review, and would be inaccurate. Insomnia is not the same as decreased sleep quality. The title should remain "Do Oral Contraceptives (OC) Affect Sleep among Women of Reproductive Age? A Systematic Review Protocol"***

Response: Thank you, we agree with this suggestion and have reverted to the initial title.